



**SAN FRANCISCO
DATA SCHOOL**

Introduction

One of the biggest challenges in getting started with data is finding the right place to start, we believe this is it. We'll explore the most fundamental aspects of data and how to work with it, without ever having to employ a data tool.

This document will help you gather a deep understanding of the world of data. We'll start with context, where you'll learn about popular data professions, tools, and processes. Next, you'll learn how to take-on the most critical undertakings with data through real-world examples. Finally, we'll reveal our most prized best practices for sharing data insights.

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Author Biography

Colby and Serge are founders of The San Francisco Data Company, an entrepreneurial endeavor to create the data education that they wish they had when they first started.

The San Francisco Data Company powers San Francisco Data School, which offers access to premium data tools and content, along with a private Slack community for anyone looking to kickstart their understanding and hands-on use of data.

Colby and Serge believe that truly understanding how to work with data changes the way people do business, live their lives, and help others.

Company Sites

<https://learn.sfdataschool.com> / <https://www.sfdataschool.com>

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LESSON 1: DATA 101

Welcome to the Data Fundamentals Handbook!

We're Colby and Serge, founders of The San Francisco Data Company. We're on an entrepreneurial endeavor to create the data education we wish we had when we first started. We power [Learn Data](#), a digital community and knowledge base for data enthusiasts

which offers access to premium data tools and content, alongside a private Slack community.

Over the next ten lessons, we'll explore the most fundamental aspects of data without ever having to employ a data tool! First, we'll start with context. Finding a straightforward explanation for some of the most basic, yet fundamental questions about data, are surprisingly difficult to find.

For instance...

Why Data, Why Now?

As in, what's with the data craze?

First, the trajectory of the volume and variety of data is absolutely massive. Volume refers to the amount of data being produced. And variety alludes to the types of data being produced. These items, together, are growing at an astounding rate.

Second, cost barriers to store this rising amount of data declines at a rapid pace overtime. And the barriers to access this data being stored are also being drastically reduced over time.

Finally, data has the ability to create a tremendous amount of value in our world. Data is, and will continue to be at the backbone of improving things in both our overall society, as well as within businesses.

So, why data, why now? Because data holds potential. And the size of that potential is growing larger and becoming easier to unlock over time.

What is Data?

This is a big, important question. It's big in that it's incredibly open-ended. And it's important in that it's so fundamental, yet so simple in composition that it often gets overlooked.

Practically speaking, data is all of the things being recorded with technology. In 1989 Tim Berners-Lee invented what ultimately becomes the World Wide Web, which has since led to the creation of countless technologies that record and store data. Like what? Let's keep moving to answer that...

Where Does the Data Come From?

Let's dive into some examples:

- Everytime a Facebook Like button is clicked, data is being recorded (i.e stored in a database). The date, user account that's done the clicking, the actual piece of content that was liked, etc.
- Everytime a Wikipedia page is authored or updated there's data being logged to account for the edit
- How about a technology like 23andMe. Each time they run a DNA test they're producing and recording data to inform you of your ancestry and genetic health risks
- Blackboard, which is geared towards helping schools connect teachers, students and parents. The sharing of grades, uploading of homework, and messages sent all produce data
- The United States Census and the surveys they conduct to help inform us on things such as population growth. Each completed survey is a recording of data

There are countless examples we could share, but hopefully through these five the intersection of data and technology is becoming evident.

Who Uses Data?

Looking at our examples again, if we abstract our thinking up one layer we see Facebook as simply an Internet company. Wikipedia is a nonprofit. 23andMe is in the biotechnology industry. Blackboard is being used by education systems. And finally, the US Census is a branch of the government.

So in just five examples we see there's a huge range with respect to who's using data – from governments, to internet companies, to nonprofits and beyond.

What is Data Used For?

Data is primarily used for three things:

1. Make better decisions
2. Build better products
3. Enrich our stories

Let's use our US Census survey as an example. This survey data is used by:

- Local governments to make decisions regarding public transportation
- National agencies to build models to better predict population trends
- Journalists to support their articles and/or documentaries

We'll stop here for now, as you'll see these questions continue to come to life through the rest of this handbook.

Book Recommendation

[Dataclysm](#), where Christian Rudder uses data to show us who we truly are

LESSON 2: DATA PROFESSIONS

In Lesson 1 ([Data 101](#)) we addressed some of data's most fundamental questions. Here, you'll explore and build acumen around prevalent Data Professions. Let's jump right in!

There are three overarching professional domains:

- Data Analytics
- Data Science
- Data Engineering

We'll first break them down one-by-one to clearly distinguish their differences, and finish with a simple exhibit of how all three generally work together to make the most of data.

Data Analytics

Often referred to as just Analytics, professionals operating in this domain are quality assuring data, building reports and dashboards, and running basic analyses to help an organization answer four high-level questions with data for the purpose of optimal decision making:

What happened?

How many users registered on our mobile app yesterday?

Why did it happen?

Which platform registered the most mobile app users yesterday, iOS or Android?

What is likely to happen?

How many mobile app users will we have by the end of the year?

What actions should we take?

What experiment should we run to better understand how to convert users faster?

Common roles in this domain generally end with "Analyst" (e.g. Data Analyst, Business Analyst, Sales Operations Analyst, Growth Marketing Analyst, Product Analyst, etc.)

Data Science

While the field of Data Science is vast, we can break it down into two simple genres: Decision-Based, and Feature-Based Data Science. Let's dig a bit deeper:

Decision-Based Data Science

This genre of Data Science is simply an extension of Data Analytics. People operating in this space instrument the techniques that help tackle overly complex problems that require advanced data-modeling and statistical techniques. Specifically, they help build sophistication around answering the last two questions from our list above (i.e. What is likely to happen / What actions should we take)

For example, the interpretation of whether or not a product experiment was successful (especially if it was tested with a small sample of customers) would typically be categorized as a Data Science effort, as it would require the use of advanced statistical and probabilistic methods.

The common role in this area of Data Science is: Data Scientist

Feature-Based Data Science

This genre of Data Science refers to the building of data products. Data products are literal product features based-on data that are consumed by a companies' external users/customers. Let's dive right into an example:

Amazon's recommendation feature, "*Customers Who Viewed This Item Also Viewed*" is a data product. Data Science folks at Amazon have built a data-model to relate transactions that have been purchased together with the item you've just selected. Amazon is using data to not only deliver this data product, but continue to make it more accurate over time.

The common role in this area of Data Science is: Machine Learning Engineer

Data Engineering

In order for any of the above to be possible (i.e. Data Analytics and Data Science), data actually has to be available. Analysts, Data Scientists, and Machine Learning Engineers need access to clean, organized, and timely data to do their work. This is where Data Engineering comes into the picture!

Sometimes referred to as Data Infrastructure or Data Architecture, Data Engineers help build the "data pipelines" that collect, store, update, and move data from one place to the next. These efforts are typically engineering-forward, in contrast to being analytical.

A common role in Data Engineering is: Data Engineer

So, in summary, folks in Data Engineering help build the infrastructure for data to be accurately collected, so that folks in Data Science and Analytics actually have data to work with!

Next, we'll discuss the Data Tools these professions use to get their work done!

Book Recommendation

[Data Science for Business: What You Need to Know about Data Mining and Data-Analytic Thinking](#)

LESSON 3: DATA TOOLS

In lesson 2, we explored the three main areas of the data profession:

- Data Analytics
- Data Science
- Data Engineering

Here, we'll cover the different Data Tools needed to start working with data. Let's get started!

Where To Start?

The vast majority of data analysis projects can be tackled by using a combination of three types of data tools:

- Spreadsheets (Excel, Google Sheets, ...)
- Databases & SQL (MySQL, PostgreSQL...)
- Business Intelligence (BI) Software (Looker, Tableau, ...)

To help us better understand why we only need these three tools, it's useful to cover the different capabilities of data tools.

Data Tool Capabilities

Each data tool can be classified by a combination of five different capabilities:

1. Storage - does the tool have the capability to keep data in it?
2. Structure - does the tool automatically provide a level of organization to data?
3. Access - does the tool allow you to view and/or extract data?
4. Manipulate - does the tool enable you to build calculations, edit values, and more?
5. Sharing - does the tool have the ability to build reports, graphs, and more?

Each data tool has its own area of specialization, and some combine multiple capabilities in a single tool.

Why Only Three Data Tools?

We strongly believe that it's best to only focus on learning spreadsheets, database + SQL, and a BI tool when first getting started to work with data for three reasons. First, as previously mentioned, the combination of these three tools can help you solve the vast majority of data analysis problems.

Second, because of their capabilities, they are the most popular tools used in data analysis, so learning these is critical for a career in data, and you can find a lot of information by searching online.

Finally, these tools make it easy to learn key principles of data analysis, without being distracted by the complexities of other tools or programming languages like Python and R. Now let's take a closer look at each type of tool and their range of capabilities.

Spreadsheets

Spreadsheets are the most popular and arguably the most powerful type of data tool in the industry because they offer the entire spectrum of capabilities in a single tool. Using Excel as an example:

- Storage - data contained within the file: the more data the larger the file
- Structure - data structured into rows, columns and individual cells
- Access - data can be accessed by opening the file
- Manipulate - Excel provides numerous mathematical and formatting functions
- Sharing - Visualizations can communicate insights, and the file itself can be shared

Databases & Structured Query Language (SQL)

Most of the world's data are stored in a database, and the most common way to access this data is through writing SQL. Storing data in a database and writing SQL to access it unlocks the entire spectrum of capabilities, with the exception of Sharing:

- Storage - data stored in a local database (e.g. laptop) or on a cloud server
- Structure - data structured into tables and schemas
- Access - data accessed via SQL code
- Manipulate - SQL can apply mathematical functions, transforms, etc.

Business Intelligence Software

Business intelligence tools are software applications that are designed primarily to analyze and report on data stored in a database. Metaphorically, BI Software sits atop a database and unlocks the sharing capability (e.g. dashboards, scheduling reports, etc.). Behind the scenes, the BI tool is writing SQL to access a database, and returning the data in its graphical user interface. BI software alone has the following capabilities:

- Access - can connect directly to a database
- Manipulate - provides drag and drop functionality to data analysis
- Sharing - easy to create beautiful visualizations

But, because BI software connects to a database the entire capability spectrum is unlocked with the added advantage of being online and capable of working with large datasets.

In summary, for tasks where data is available in a CSV or Excel file, a spreadsheet tool is all you need for your analysis. For data sets that live in a database, the combination of SQL plus a BI tool should be your tools of choice.

Next, we'll take a look at different data files and data types, and the impact they have on data analysis.

Book Recommendation

[Data Smart: Using Data Science to Transform Information into Insight](#)

LESSON 4: DATA FILES, FORMATS, AND TYPES

In lesson 3, we learned that there are three main tools that can cover the vast majority of data analysis projects:

- Spreadsheets
- Databases & SQL
- BI Software

Here, we'll introduce the concepts of Data Files, Formats, and Types. Let's jump in!

Data Files

Think file extension when you hear Data File. In other words, the way in which a file is saved – which has an impact on how different software applications access and consume the data within it.

For example, Excel files are saved with the extension .xlsx. The .xlsx data file is Excel's way of saving data in a particular structure, so the Excel program knows how to read and process the data stored in it. The most common Data Files are :

- CSV - Comma Separated Values, unformatted text or numbers delimited by commas
- TSV - Tab Separated Values, unformatted text or numbers delimited by tabs
- TXT - Standard text document that contains an unformatted text document
- DOCX - Microsoft Word specific extension for a formatted text document
- XLSX - Microsoft Excel specific extension for a formatted spreadsheet

Data Formats

Data formats refer to how the data fields are formatted, or structured, within a file itself. In the previous section, we compared CSV to TSV file formats. Both of these file formats save data in a flat way, meaning there aren't any nested relationships between data records within the file. For this reason you'll hear CSV and TSV files being referred to as flat files that have a fixed structure.

In cases where the data records contain more complex relationships, we need to store the data in a more flexible format that allows for these relationships to be maintained.

For example, what if we wanted to store data for someone's name and type of car:

Name	Car
Serge	Ford
Colby	Saturn

But, what if one person has two cars? Would we add a third column (see table below), and leave the other person's cell blank (an inefficient use of space)? What if someone had five cars?

Name	Car	Second Car
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Serge	Ford	
Colby	Saturn	Honda

This is where semi-structured/nested data formats come into play, and a popular data format for this type of use case is called JSON (JavaScript Object Notation). JSON is made up of what we call key - value pairs, the names on the left being the keys, and the fields on the right hand side being the values, all being enveloped by curly braces. For each key, there can be any number of different values, without having to add columns.

The above example in JSON would look like this:

```
[
  {"Name": "Serge", "Car": ["Ford"]},
  {"Name": "Colby", "Car": ["Saturn", "Honda"]}
]
```

The point here isn't to be an expert on decoding JSON, rather it's to simply know there's various forms of storing and structuring data. For instance, in addition to JSON some other common formats are Avro, ORC and Parquet.

Data Types

Data types are simply categories assigned by a data tool or programming language that provide information about the range of possibilities for the values that data can have. There are many different data types in the industry, but the three main categories are:

- Text - typically referred to as a string, and can have a variable (e.g. a comment box on a website) or fixed size (e.g. 2-Letter Country Code)
- Numbers - different types include integer (round numbers), numeric (decimal values) and boolean (0 or 1)
- Dates - describe dates in time, often stored in a "year-month-day" format

Data types have a big impact on data tool functionality and performance, and can lead to incorrect interpretation of results in some cases. For instance:

- Financial transaction data saved as Integers instead of Numeric would round to the nearest dollar
- Dates saved as a Timestamp without knowing timezone would be impossible to convert

Now that you have an understanding of the different ways we can store data, next, we'll take a look at cleaning and organizing data!

Book Recommendation

[Confident Data Skills: Master the Fundamentals of Working with Data and Supercharge your Career](#)

LESSON 5: CLEAN AND ORGANIZE DATA

In lesson 4 ([Data Files and Types](#)) we walked through the most regularly used data files, and untangled the reasoning behind why there's more than one. We also revealed common data types, and exemplified their importance through real-world experiences.

Here, we'll change gears, and move from data theory to practical data knowledge and techniques. We believe there are three critical areas here, and the first is to [Clean and Organize](#) data. Here's what we mean...

Overview

In a perfect data world, you'd have the luxury to spend the bulk of your energy concentrated on identifying patterns, building beautiful charts, further enhancing a predictive model, and sharing the most exciting of insights. Basically, you'd be doing actual analysis (the fun part of working with data).

However, in the [vast majority](#) of projects data practitioners spend the [vast majority](#) of their time simply preparing (cleaning and organizing) data for analysis. In fact, working with data is commonly associated with the Pareto Principle (80:20 Rule), in that you'll spend 80% of your time preparing data and 20% performing analysis – it's unevenly distributed.

But, it's for good reason! And it's because: **Garbage In = Garbage Out**. An analysis won't be very effective if you're working with inaccurate data in the first place. You'll have a really hard time getting clarity on typical customer behavior if your customer database is missing 50% of its records.

The point is, data quality is directly correlated to analysis quality. There's a motto in the data profession that "quality data beats fancy algorithms", implying that sophistication and fancy math won't fix low quality data. Here's a few specific examples of Garbage In (many times known as "dirty data") that [we've personally experienced](#):

- Global finance data was supposed to be converted to a single currency, yet upon further digging we discovered many entries were still being stored in their native currency (e.g. Mexican Pesos weren't being converted to USD)
- Customer support data was supposed to be unique (e.g. we should receive one entry for every customer phone call). However, upon further digging we discovered that duplicate entries were finding their way into our systems
- Product data was supposed to be up-to-date in our analysis of an experiment, yet upon further digging we discovered that an issue in our database was causing many records to never get created in the first place

In all scenarios we would have shared an inaccurate analysis. Not because our analysis was wrong, but simply due to the use of "dirty data". Hopefully it's becoming crystal clear as to just how important this is.

Framework

Below is a general framework/checklist you can use in your daily data extravaganzas to verify whether or not you're working with clean data:

Valid

Is the data accessible, up-to-date, and properly stored?

If you need access to time-series data with timestamp and timezone, but your data is being rounded to the day this would be invalid data

Accurate

Is the data actually correct?

If you realize that your sales team has been "fudging the numbers" on their pipeline forecast, you'd be working with inaccurate data

Complete

Is there any data missing and/or duplicated?

If you search for customer invoice data with a specific tag, but invoices have been improperly tagged, you'd be working with incomplete data

Uniform

Is the data inputted the same way every time?

If you notice that customer phone numbers are stored in various ways (e.g. some with dashes and some without, some with country codes and some without, etc.), you'd be working with inconsistent data

So, in summary, cleaning and organizing data is where analytic professionals spend most of their time. This is due to data being so easily prone to error, in combination with the importance of analyzing clean data (i.e. Garbage In = Garbage Out).

Next, we'll discuss our second practical area, which is the joining of multiple datasets!

Bonus Tip

Before you make any changes to a dataset, make a copy first. Then, while you work through the framework above track all your changes. This way you can then clearly articulate to others exactly what's changed, and show them the "before vs. after".

Book Recommendation

[How to Lie with Statistics](#)

LESSON 6: JOINING DATA TOGETHER

In lesson 5, we shared the first critical area for practically working with and understanding data – Clean and Organize. Analytic professionals spend most of their time preparing data for analysis, because as the saying goes, "Garbage In, Garbage Out". Making sure data is valid, accurate, complete, and uniform is one of the most important things you can do with data.

Here, we'll uncover the second critical area for practically working with and understanding data – Joining Data Together. The proper way to address this topic is Relational Data Modeling, and let's work through an analogy to help bring this concept home – have you ever heard the "Dem Bones" melody?

If not, it goes something like this:

"Toe bone connected to the foot bone. Foot bone connected to the heel bone. Heel bone connected to the ankle bone. Ankle bone connected to the shin bone..."

The point is, each bone is connected (i.e. relational) to another bone, thus enabling the entire skeletal system to properly connect and work together. Well, there's an established way to build a data model in this same way, and it's known as a Relational Data Model.

In a Relational Data Model, each dataset is built so that it can also connect to another, or multiple other datasets. And ultimately, they can all join together! Before we dive into some examples, let's first emphasize the reasoning behind why datasets get separated in the first place.

Imagine yourself building a web application where users can:

- Register on your website
- Purchase a product on your website
- Fill out and submit a support form on your website for help

Now, imagine you're forced to store all the data that's produced from these actions into a spreadsheet. How would you store all the data? Does it make sense to try and store everything in one spreadsheet tab?

If so, maybe it would look something like this for a single user:

User ID	Register Date	Purchase Date	Support Date
C072X	2019-01-01	2019-01-15	2019-01-16

Seems simple, right? But, what if our user makes another purchase? Do you add another row and just leave the other cells blank, like so:

User ID	Register Date	Purchase Date	Support Date
C072X	2019-01-01	2019-01-15	2019-01-16
C072X		2019-01-20	

Or, is it better to just override our original purchase date with the most recent?

What if you decide to start selling more than one product, allow users to do more actions on your website, and want to see more data than just the date for help requests – do you just keep adding new columns?

Hopefully you're starting to see the lack of flexibility inherently created with a single "master" table. Not to mention the headaches you'd get from trying to clean the data, gosh forbid any errors found their way in (trust us, they always do).

It's best to separate the data and architect a link between them. Each dataset lives in its own domain, and can thus be analyzed individually. Yet, the architected link provides the power to analyze holistically when necessary. For instance, we could have three tabs in our spreadsheet, and each tab/dataset can join together via the User ID (the architected link):

Registrations

Payments

Support Inquiries

User ID	Register Date	User ID	Paid Date	User ID	Support Date
C072X	2019-01-01	C072X	2019-01-15	C072X	2019-01-16

This way you can simply append more data, and/or expand the number of fields (i.e. columns) without having to worry about leaving cells empty or deleting historical data. You also have the ability to create a “master” table if you so choose by joining the data together via the link (i.e. User ID) – this is a Relational Data Model.

To close out here, have you ever used the VLOOKUP formula in a spreadsheet? If so, congratulations – you’ve joined data together and leveraged the concept of a Relational Data Model! In SQL, joining data together is performed via a JOIN statement. And BI Software leverages SQL, and is thus doing the same thing behind the scenes (i.e BI Software writes SQL to a database). Next, we’ll address our final area for practically working with data – Sort, Filter, and Segment!

Book Recommendation

[Lean Analytics: Use Data to Build a Better Startup Faster](#)

LESSON 7: SORT, FILTER, AND SEGMENT DATA

In lesson 6, we shared the second critical area for practically working with and understanding data: Joining Data Together. Relational data models are powerful. Each dataset lives in its own domain, yet an architected link provides the power to analyze holistically.

Here, we’ll share the third and final area for practically working with and understanding data: Sort, Filter and Segment. Let’s jump right in!

Sort

The beauty of sorting data is that it’s a fairly simple concept to grasp, and can provide insights incredibly fast. Sorting is many times referred to as ordering, and there’s two main concepts to internalize here:

Ascending sort is when things go from **low to high**. Here’s a few examples of ascending sort:

- 1, 2, 3, 4, 5
- Amanda, Colby, Marissa, Timber (i.e. alphabetical)
- 2019-01-01, 2019-02-01, 2019-03-01 (i.e. old dates to new)

Descending sort is when things go from **high to low**. Here’s are same examples, but now in descending sort:

- 5, 4, 3, 2, 1
- Timber, Marissa, Colby, Amanda
- 2019-03-01, 2019-02-01, 2019-01-01

Sorting is a really quick way to see the highs and lows in a dataset. In spreadsheet tools there’s legitimately a feature dedicated to sorting. In SQL this is done by the ORDER BY

operator. And in BI Software, sorting is typically performed by clicking on the field of interest.

Filter

Filtering data is the act of purposefully choosing to include and/or exclude certain data points. Here's a few examples of filtering data:

- You have 24-months of customer support data, but only care to see the prior 12-months. *You'd thus want to exclude the oldest 12-months via a filter*
- You have customer support data by country, but only want to analyze interactions in North America. *You'd thus want to exclude all other countries via filter*

Filtering, like sorting, is a fairly straightforward concept (i.e. to include or exclude). The concept we see most folks stumble on, is filtering for multiple criteria. When looking to join multiple filters together you have two forms of logic:

AND – data must meet all filter criteria to be included/excluded. *Carrying on with our example above, we want to see the prior 12-months AND requests in North America (i.e. only the data that meets both conditions)*

OR – data must meet at least one filter criteria to be included/excluded. *Carrying on with our example above, we want to see the prior 12-months OR requests in North America (i.e. we'll see North America requests older than 12-months + other country requests within the prior 12-months)*

In spreadsheet tools there's a feature dedicated to filtering, along with formulas for more complex AND/OR logic. In SQL filtering is done via a WHERE clause, and use of the AND/OR operators. And in BI Software, filtering is typically a built-in feature directly within the UI.

Segment

Datasets typically contain a lot of variability and depth, making them hard to process. Segmentation is powerful because it enables you to categorize/label data points to their core theme, thus making your dataset easier to interpret. Let's jump into an example:

Pretend you work for a business that sells a product to every country in the world (as of this writing, there are 195 countries in the world). It would be a difficult exercise to analyze and extract insights for all 195 countries. However, we could segment each country to its appropriate content, and then have only 7 areas to analyze.

We've used segmentation to help our companies understand the:

- Time to customer purchase (1 Day, 2-3 Days, 4-7 Days, ...)
- Volume of help tickets by type (Billing, How To, Product Bug, ...)
- Style of device (Mobile, Desktop, Gablet, ...)

In spreadsheet tools segmentation is best done via pivot tables. In SQL, segmentation is done via a GROUP BY clause. And in BI Software, segmentation is typically a built-in feature directly within the UI (many times titled "Pivot").

Sorting, filtering, and segmenting data are three powerful techniques for simplifying a dataset and making it easier to understand the things that matter most. Next, we'll change gears and tackle our first data measurement principle: Central Tendency!

Book Recommendation

[The Visual Display of Quantitative Information](#)

LESSON 8: CENTRAL TENDENCY

In lesson 7, we shared the third and final area for practically working with and understanding data – [Sort, Filter, and Segment Data](#). Short sentence here...

Here, we'll change gears and move from practical work to data measurement techniques. We believe there's two that everyone interested in working with data should master, and the first is [Central Tendency](#).

Let's dive in with a trick question!

Imagine you have \$1,000,000 in an investment account (one day, right). Over the course of two years you generate the following returns:

- Start of Year 1 ➡ End of Year 1: +100%
- Start of Year 2 ➡ End of Year 2: -50%

What is your average annual return? Well, let's do the math: $(100\% - 50\%) \div 2 = +25\%$

Now, let's look at the actual movement of money for each period:

- Start of Year 1: \$1M ➡ +100% ➡ End of Year 1: \$2M
- Start of Year 2: \$2M ➡ -50% ➡ End of Year 2: \$1M

So, you started Year 1 with \$1,000,000 and ended Year 2 with \$1,000,000 (i.e. \$0 of growth)...but your average annual return is +25%?

There's two takeaways from our trick question:

First, always make sure to look at the underlying data that's producing a descriptive statistic. Sometimes average-oriented metrics don't align with what's actually occurring in the real world. If access to the data isn't an option (either the dataset is too large, or simply inaccessible), it's helpful to at least inquire about a statistic's composition (more on this below) and the data source.

Second, we highly recommend to start thinking of "the average" as [just a way of articulating the central tendency](#) in a dataset, and not as a literal mathematical computation which most often is associated with the [Mean](#). In fact, there are two other ways to calculate "the average" that we recommend: [Median](#) and [Mode](#).

Let's line-up another example to reiterate why the typical "average" (i.e. Mean) [is normally a fine way to define the center, but not always](#). Below we have three tables, with each having the exact same data. The very last row in each table calculates a measure of central tendency:

- Mean - sum all values in a dataset, then divide by the number of values
- Median - after a dataset is sorted, this is the middle number (i.e. 50th percentile)
- Mode - the number that repeats most often in a dataset

Name	Weight
------	--------

Name	Weight
------	--------

Name	Weight
------	--------

Timber	500	Timber	500	Timber	500
Kygo	75	Kygo	75	Kygo	75
Rooby	50	Rooby	50	Rooby	50
Aslan	50	Aslan	50	Aslan	50
Rome	65	Rome	65	Rome	65
Burton	70	Burton	70	Burton	70
Mean	135	Median	68	Mode	50

Here's the takeaway – the Mean is sensitive to outliers. Notice in our first table that the Mean (135) is being pulled up due to the outlier weight of Timber (500). No other value in our dataset is even close to 135. Outliers in data can distort/skew figures, and not provide a true representation of what is really happening.

To summarize our recommendation when working to define the central point in a dataset, always pull all three measures of central tendency (Mean, Median, and Mode). Not only will this make your analysis look fancy and trustworthy, but it will provide you and others with legitimate insight into what "the average" actually is and whether or not there's any distortion/skew in the underlying data.

Next, we'll dive into our second data measurement principle to master: Growth and Normalization.

Book Recommendation

[Naked Statistics: Stripping the Dread from the Data](#)

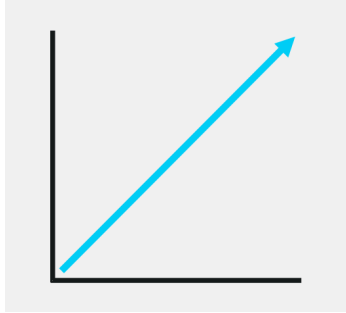
LESSON 9: GROWTH AND NORMALIZATION

In lesson 8, we shared the first critical data measurement technique to master – Central Tendency. Here, we'll uncover the second measurement area to master – Growth and Normalization.

Nearly everything we do as a species revolves around change— is the company we work for getting larger, favorite sports teams getting stronger, waist size getting smaller...you get the point. It's at the crux of most things we do. This is especially true at a professional level. For instance:

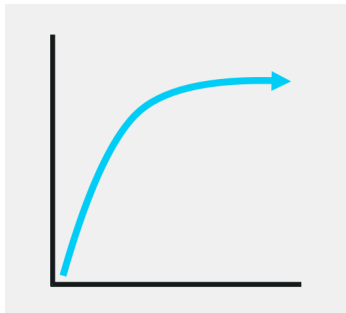
- Sales Manager: is our team's average deal size increasing over time?
- Financial Analyst: are company revenues increasing?
- Customer Support Agent: are we maintaining strong customer satisfaction scores?

Although different, they all share the common thread of articulating change (i.e. growth, which could be positive or negative). At a high-level, there's three primary ways something grows overtime: Linear, Logarithmic, or Exponential. Here's an image and example depicting each:



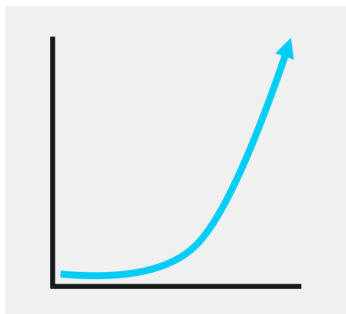
Linear results grow at a consistent amount in each step.

For instance, growth in one's paycheck while working at an hourly job. Income will increase directly alongside the number of hours worked.



Logarithmic results occur rapidly with significant advancement, yet dwindle as time continues on.

For instance, growth in market share for a local grocery store. Upon opening, advancements will occur quickly as customers discover and start relying on it. Yet, as time goes on it'll begin to plateau for that single location.



Exponential results slowly build on top of each other, and as time goes on, the advancement from one step to the next continually increases.

For instance, growth in one's 401K Retirement Account. There's a small amount of money starting out, yet as time goes on the impact of compounding takes place and produces large advancements.

In terms of actually measuring change there's two main areas to discuss – Number to Number, and Percent to Percent. Let's dive into an example of each:

Number to Number

Pretend you just got a huge raise and went from making \$50k per year, to \$150k per year – below are three different ways you could accurately articulate this change:

Starting #	Current #	Absolute	Percent	Multiple
\$50k	\$150k	+\$100k	200%	3x

Take note that 200% is not equal to 2x (this is something we see done wrong all the time).

Percent to Percent

Pretend you refinance your mortgage and go from having a 6% interest rate, to 4.6% – Below are three different ways you could accurately articulate this change:

Starting %	Current %	Percent Point	Percent	Basis Point
6.0%	4.6%	-1.4	-23%	-140

Take note that a -23% drop is equivalent to a -1.4 Percentage Point drop in this example (we see Percentage Point get confused with Percent all the time).

Just like in our previous lesson's recommendation (show all measures of central tendency), it's best to see all metrics of change, all of the time. In doubt better to give too much information than less.

Normalization

The final thing to know about growth is normalization – are the things being measured the same, and/or is the reasoning for why things have changed sound. For instance:

- Suppose you're managing a sales team and working hard to increase your team's average sales amount. If it has grown, is it because of all the coaching, strategizing and experiments implemented, or is it simply because the business raised its prices?
- Is 25% growth in a stock price good, if competitor stocks all grew by 125%?
- Is it fair to measure a 31-Day Month vs. 30-Day Month vs. 28-Day Month?

Don't get too hung up on the details of how to calculate each measurement of change, and normalize to perfection. Instead, just try to fully internalize the fact that: there's three macro-level ways in which growth occurs, there's more than one right way to calculate change, and that change is dependent on the context surrounding it.

However, if you want to get hung up on details, check out our "Additional Recommended Reading" list below! Next, we'll talk through how to best share data insights.

Additional Recommended Reading

[If You Only Learn One Thing in Working with Data, Learn This](#)

[The Difference Between: Absolute, Basis Point, Multiple, Percent, and Percentage Point](#)

Book Recommendation

[A Mind for Numbers: How to Excel at Math and Science \(Even If You Flunked Algebra\)](#)

LESSON 10: STORYTELLING WITH DATA

In lesson 9, we shared the second critical area for data measurement: Growth and Normalization – there's three macro-level ways in which growth occurs, there's more than one right way to calculate change, and that change is dependent on the context surrounding it.

Here, we'll provide our two favorite frameworks for helping others get the most out of your data analysis work! The first framework is for:

Presenting Spreadsheet Data

Step 1: Describe each field, and walk through a single row of data

All too often, presenters of a spreadsheet jump into "the weeds" – details that are important and make sense to them, but typically not important and confusing to others. Take the time to explain what each field/column represents:

- Column 1 (e.g. Date) denotes the date in PST
- Column 2 (e.g. # of Sales) total number of sales recorded, excluding refunds
- Column 3...

Once you have everyone informed on the fields, walk through a single row of data and tell the story for that row. This will help you solidify understanding of what's being displayed, but more importantly help you segue to the next two steps.

Step 2: Validation of Data Integrity and/or Material Annotations

Skepticism on the validity of data is common, and rightly so – sometimes it's hard to get your hands on a perfectly "clean" dataset. One way to ease the tension here is to put your validation efforts on the table:

- Where was the data sourced?
- Was there any clean-up necessary, and if so, what did that consist of?
- What context surrounds the dataset that's worthy of an annotation?

Pick no more than three of the most important annotations you'd like to make, and share them. If you're working with a smaller, and knowingly "perfect" dataset, then possibly you could skip this step. However, in most cases we've found this to be an incredibly powerful way of building trust and showing that you're well-attuned to the dataset.

Step 3: Insights + Recommendations

Now that you've got everyone on the same page as to what's in the spreadsheet (#1), there's trust in the data being presented and all are aware of any material context (#2), it's time to deliver the goods: insights and recommendations. This could come in a variety of ways:

- What's the most interesting thing you noticed?
- Is there anything particularly great, or negative happening?
- Where should we go from here?

Pick no more than three of the most important annotations you'd like to make, and share them. If you're working with a smaller, and knowingly "perfect" dataset, then possibly you could skip this step. However, in most cases we've found this to be an incredibly powerful way of building trust and showing that you're well-attuned to the dataset.

The second framework is for:

Presenting a Data Visualization

Step 1: Take the time to explain what's being visualized

All too often, presenters of a data visualization jump-in to "the weeds" – details that are important and make sense to them, but typically not important and confusing to others. Take the time to explain what's being visualized:

- What's on the x-axis: at what increment / in what order / starting and ending at
- What's on the y-axis: at what increment / in what order / starting and ending at
- High-level description of what's being visualized (*i.e. So, what you're looking at is...*)

Step 2, and **Step 3** are the same as above! Once you have everyone informed as to what they're looking at, provide relevant context and then deliver the most fascinating insights and recommendations!

Pro Tip

The above framework can also be used for sending an Email. However, we recommend reversing the order (Insights ➡ Annotation ➡ Guide), and keeping it short.

In summary, there is no right way to present a spreadsheet, or data visualization. Possibly you'd prefer to share context after the insights and recommendations have been shared. Or, maybe you don't care to receive validation on the integrity of a dataset – so be it. The above is simply a cadence we've found to help set a strong foundation for the insights and recommendations provided.

Thank you for reading the Data Fundamentals Handbook!

– Colby and Serge

Book Recommendation

[Storytelling with Data: A Data Visualization Guide for Business Professional](#)

Quiz

Question 1: What is data used for?

- A. Make better decisions
- B. Build better products
- C. Enrich our stories
- D. All of the above

Question 2: Which one of these is not one of the main fields in the data profession?

- A. Data Analytics
- B. Data Science
- C. Data Products
- D. Data Engineering

Question 3: What type of data tool offers the entire spectrum of data tool capabilities?

- A. Spreadsheets
- B. Database + SQL
- C. Business Intelligence Software
- D. They all have the same capabilities

Question 4: Which data file format represents a file of unformatted text separated by tabs?

- A. CSV
- B. JSON
- C. TXT
- D. TSV

Question 5: For most data projects, what % of time is spent cleaning data vs doing analysis?

- A. 90:10
- B. 80:20
- C. 50:50
- D. 30:70

Question 6: Which function in Excel or Google Sheets can be used to join data together?

- A. SUM
- B. COUNT
- C. VLOOKUP
- D. JOIN

Question 7: Which Excel or Google Sheets feature allows you to easily segment your data

- A. Pivot table
- B. Sort function
- C. Filter function
- D. Data validation

Question 8: Which function gives a good representation of the “average” value in a dataset when the dataset contains outliers?

- A. Mean
- B. Median
- C. Mode
- D. All of the above

Question 9: Which type of growth curve has growth where advancement from one step to the next continually increases?

- A. Linear
- B. Logarithmic
- C. Exponential
- D. All of the above

Question 10: What is the best way to share the results of your analysis?

- A. Email
- B. Slide deck with visualizations

- C. Detailed report
- D. All options are valid, best way depends on circumstances

Quiz Answer Key:

- 1. D
- 2. C
- 3. A
- 4. D
- 5. B
- 6. C
- 7. A
- 8. B
- 9. C
- 10. D